Solutions for Homework 1

1. Consider a disk with a sector size 1024 bytes, 1000 tracks per surface, 100 sectors per track, 3 double-sided platters, and average seek time of 16 milliseconds.
   a) What is the capacity of the disk?

   Answer: Disk capacity = 3 × 2 × 1000 × 100 × 1024 bytes

   b) What is the capacity of a cylinder?

   Answer: Cylinder capacity = 3 × 2 × 100 × 1024 bytes

   c) If the disk platters rotate at 3000 rpm, what is the average rotational delay?

   Answer: Avg rotational delay = $\frac{1}{2} \times \frac{1}{3000} \times 60 \times 1000$ = 10 milliseconds

   d) What is the block transfer time if one block contains 3 sectors?

   Answer: btt = $\frac{Maximal\ rotational\ delay}{\#\ of\ blocks\ in\ a\ track} = \frac{2 \times 10}{100/3} = \frac{3}{5}$ = 0.6 milliseconds

2. Given two sorted sequential files A and B with n (number of records) = 100000 and R=500 bytes each, we want to create an intersection file. Assume that 60% of the records are in common and the available memory for this operation is 10 Megabytes. Calculate a timing estimate for deriving and writing the intersection file. Use s = 16 milliseconds, r = 8.3 milliseconds, btt = 0.84 milliseconds, B = 2000 bytes.

Answer: Each file (A or B) has 100000 × 500 ≅ 50 Megabytes. Since our RAM has 10 Megabytes, it is not possible to read files A and B entirely to memory. So we read *a* pages from file A, and *b* pages from file B, compute the intersection of the pages in RAM, then write the common records to *c* pages in RAM. When the *c* pages containing the results become full, we write the *c* pages to disk. Totally we need to read A and B files entirely just once since these two files are sorted, and write the intersection file once to disk.

In memory, we have $\frac{10\ Megabytes}{2000\ bytes} \cong \frac{10000000}{2000} = 5000$ pages.

So, $a + b + c = 5000$

In each file (A or B) we have $\frac{100000 \times 500}{2000} = 25000$ pages.

In the intersection file we have $\frac{100000 \times 0.60 \times 500}{2000} = 15000$ pages.

The time required to form the intersection file is equal to $\left\lceil \frac{25000}{a} \right\rceil \times (s + r + a \times btt) +$

$$\left\lceil \frac{25000}{b} \right\rceil \times (s + r + b \times btt) +$$
$$\left\lceil \frac{15000}{c} \right\rceil \times (s + r + c \times btt)$$

where the first component is the time required to read file A, the second component is the time required to read file B, and the third component is the time required to write the intersection file.

We can use any values for *a*, *b*, and *c*. If we choose *a* = *b* = 2000 and *c* = 1000 then, we need approximately 57 seconds to form the intersection file.

3. Given the following s = 16 milliseconds, r = 8.3 milliseconds, btt = 0.84 milliseconds, B = 2000 bytes. A 20 Megabytes file is stored on this disk.

   a) If this file is a pile file, calculate time to fetch a certain record.

   Answer: Avg time required to fetch a certain record = $s + r + \frac{b}{2} \times btt$
   where b is the number of pages in the file. For this question $b \cong \frac{20000000}{2000}$ = 10000

   Time required = $16 + 8.3 + \frac{10000}{2} \times 0.84 = 4224.3\ milliseconds \cong 4 seconds$

   b) If this file is a sorted sequential file (no overflow area), calculate time to fetch a certain record.

   Answer: Time required to fetch a certain record = $(\log_2 b) \times (s + r + btt) \cong 335.2\ milliseconds\ \cong 0.3\ seconds$

   c) For each of the above cases, calculate time to fetch 100000 records.

   Answer: If pile file is used, time to fetch 100000 records = 100000 × 4 = 400000 seconds ≅ 111.1 hours

   If sequential file is used, time to fetch 100000 records = 100000 × 0.3 = 30000 seconds ≅ 8.3 hours

   d) Which file structure is more suitable to fetch 100000 records? Explain.

   Answer: Since we need less time in sequential file structure, sequential file structure is more suitable to fetch 100000 records.